# Analytical Gel

`gel`

### WHAT THIS IS FOR

Checking that a PCR produced a product and that it is the right size. You are **not purifying** — nothing is recovered from this gel.

Slabs are **1% agarose in 1× TAE**, cut from the pre-made gels in the fridge.

### PREPARE EACH SAMPLE

**8** loading dye (tube marked *load*)
**3** PCR product
Mix, quick spin.

**Marker, one per section:** 8 µL loading dye into a tube of marker (yellow), mix, spin.

### SET UP AND LOAD

1. Cut a slab with enough wells for every sample, the marker, and a few spare.
2. Put it in the rig. Fill with **1× TAE** until it is just covered.
3. Brace the gel with the plastic wedge so it cannot float.
4. Write the sample order on a paper strip; line the tubes up to match.
5. Load **9 µL** per well with a P20.

### RUN

**Run to red.** DNA is negative and moves to the red electrode, so load at the black end.

Run at **175 V** for **~10 min**, or until the blue front is **2/3–3/4 down the gel**.

> **Watch the gel, not the clock.** 175 V for 10 min has not been re-measured on the B144 power supply and rigs. The dye front is the real endpoint. Tell your supervisor what actually worked.

### IMAGE

1. Move the gel to the imager and lay the label strip above it.
2. Photograph it **with your phone through the orange filter**.
3. Name the file by date and time, then upload it.

### READING IT

– Smaller fragments move faster.
– A **library** gives a smear, not a sharp band — its members differ in length.
– Loading dye does two jobs: glycerol sinks the sample into the well, and the colour tracks the run. It does not bind DNA.
– Load below the full volume on purpose — 9 µL of 11 µL — so you do not draw an air gap.
– Do not puncture a well. The sample runs out the bottom; use another lane.